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A realist synthesis of the implementation of digital health strategies for enhanced health outcomes in Sub-Saharan Africa

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ABSTRACT

Many sub-Saharan African (SSA) countries have developed national Digital Health (DH) strategies. However, there is significant variability in how these strategies are implemented across different regions, especially in resource-constrained settings. This paper conducts a realist synthesis to explain *how, for whom*, and *under what conditions* the implementation of DH strategies facilitates sustainable and equitable health outcomes in SSA. A realist review of 30 articles (2014-2025) from Scopus, Google Scholar, and PubMed was conducted. Using Context-Mechanism-Outcome configurations, the analysis identified the underlying causal mechanisms necessary for effective implementation. The synthesis refined seven interconnected programme theories, clustered into three functional categories. 1) Foundational enabling conditions encompassing coordinated governance, strategic institutionalization, and a legal trust framework to establish legitimacy and safety. 2) Technical and human enablers encompassing infrastructure reliability and socio-technical capability-building with peer validation to create operational capacity. 3) Adaptive mechanisms encompassing participatory co-design and evidence-informed adaptation to ensure contextual fit and sustainable evolution. This review suggests that effective, sustainable, and equitable integration of DH is not achieved through isolated interventions, but through a synergistic ecosystem of seven interconnected mechanisms. By reframing the challenge for policymakers and implementers from asking what technology should be deployed to which foundational, enabling, and adaptive mechanisms must be activated for success, the review offers a critical shift in perspective that advances both policy and practice.

1. Introduction

Digital Health (eHealth) strategies are critical for guiding the integration of information and communication technologies (ICTs) into healthcare systems (World Health Organization [WHO], 2005; 2021). These strategies, whether used as standalones or in combination with other health and national policies, influence critical aspects such as resource allocation, infrastructure development, capacity building, and stakeholder engagement (WHO, 2012; 2021). Within the 47 member states of the WHO African Region, 36 have developed such national strategies or related ICT health policies, with the majority located in Sub-Saharan Africa (SSA) (WHO, 2014). Despite this widespread policy adoption, significant disparities in implementation progress and outcome remains, with only about one-third of these states demonstrating notable advancement. Countries such as South Africa, Kenya, Tanzania, Ghana, and Nigeria are often cited as leaders, having established relatively mature digital health (DH) ecosystems characterised by stronger governance, infrastructure, and policy integration (WHO, 2024).

At the core of their ecosystems, each country's DH strategy aims to address persistent healthcare challenges by leveraging ICTs. Faced with significant health disparities between rural and urban areas, a shortage of healthcare professionals, and a high dual burden of communicable and non-communicable diseases, these nations use DH solutions to bridge geographical and resource gaps (Maphumulo & Bhengu 2019; Modjadji 2021). ICTs enable countries to extend the reach of healthcare services, especially to the marginalised population, improve health-related data management for better decision-making, and enhance efficiency of service delivery, thereby accelerating progress towards Universal Healthcare Coverage (UHC) (Karamagi et al., 2022; Ndlovu et al., 2021). However, despite progress made, efforts to achieve sustainable, equitable, and scalable health outcomes continue to be hampered by several challenges (Alunyu et al., 2024; Christmals & Aidam, 2020; Scott & Mars, 2015).

These challenges range from inconsistencies in regulatory compliance and weak governance to a lack of sufficient accountability

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mechanisms and transparency and technical limitations (Alunyu et al., 2024; Attaran, 2022; Granja et al., 2018; Ogugua et al., 2024). The implications of these challenges have far-reaching consequences which prevent DH from reaching its full potential, often leading to wasted resources and increased inequity. For instance, weak governance and inconsistent regulations lead to a collection of disconnected DH projects rather than a unified national system. This has several negative effects such as siloed data systems that make it difficult to seamlessly share patient-related data information, resulting in duplicative efforts and delays in care (Mahomed et al., 2017; Senyoni, 2020). This in turn widens equity gaps, ultimately hindering rather than enabling the journey toward UHC under Sustainable Development Goal (SDG) 3 (Chuma & Sibiya, 2022; Spicer et al., 2020).

Overcoming these challenges requires that SSA governments critically evaluate their implementation efforts, thereby ensuring their national DH strategies or related ICT health policies can effectively leverage ICTs to meet health targets and drive meaningful and sustainable improvements in health care (WHO, 2016; WHO, 2021). Within this paper, the phrase "effective implementation" of DH strategies is conceptualised as the successful adaptation and execution of policy instruments and strategic actions designed to facilitate the integration of ICTs into healthcare workflows (Al Meslamani, 2023; Greenhalgh et al., 2017; Mehl & Labrique, 2014). Achieving effective implementation requires a coordinated effort from multiple stakeholders and unfold through several interconnected processes (Ansah et al., 2024; Ogugua et al., 2024). Key stakeholders include but not limited to agencies such as the WHO, governmental and regulatory authorities, healthcare providers, healthcare practitioners, and patient's advocacy groups, each contributing distinct yet complementary roles.

First, at a strategic level, the development and translation of DH strategies or ICT health policies can occur at various tiers: international, continental, or national (Ansah et al., 2024; Ogugua et al., 2024). At the international and continental levels, organizations such as the African Union (AU), the WHO Regional Office for Africa (WHO-AFRO), and International Telecommunication Union (ITU) have all developed and oversee several DH strategies and ICT policy frameworks. Secondly, Ministries of Health in individual countries in SSA also formulate, enforce, and oversee their DH strategies and policy frameworks informed by the internation and continental policies and strategy frameworks. Third, healthcare institutions, whether private or public, operationalize policy instruments and strategic actions within clinical settings. Finally, advocacy groups ensure alignment with the needs and priorities of patients within a specific context.

Therefore, understanding the interplay among these stakeholders and the governance mechanisms that structure their interactions is critical for ensuring effective implementation and long-term sustainability of DH strategies. However, authors like Odoch et al. (2022), Otorkpa et al. (2024), and Ansah et al. (2024) argue that this is not well-understood. More especially when it comes to tailoring and aligning international, continental, or regional policy instruments and individual country's strategic actions (Alunyu et al., 2024). This is particularly important considering that at the operational level (healthcare institutions) in varied settings, DH strategies are implemented through strategic actions such as integrating DH interventions such as mobile health (mHealth) apps, telehealth, and electronic health record (EHR) systems into healthcare workflows to improve patient care and operational efficiency (Achieng & Ruhode, 2023; Karamagi et al., 2022; Olu et al., 2019). In essence, effective implementation of DH strategies depends on the alignment of appropriate policy instruments such as regulatory frameworks, financing models and capacity building mechanisms, with targeted strategic actions that translate these instruments into measurable outcomes.

Against this background, this paper examines the critical *role of effective implementation of DH strategies and related policies in facilitating the integration of DH technologies into healthcare workflows, with the goal of enhancing service delivery and promoting sustainable and equitable health outcomes in SSA countries*. The underpinning research question is what contextual factors, and causal mechanisms influence effective implementation of DH strategies and related health policies in facilitating sustainable and equitable DH outcomes across SSA. The main objective of the paper is to conduct a realist synthesis to establish the following: (i) the mechanisms through which effective implementation of DH strategies facilitate adequate integration of DH technologies into healthcare workflows to improve service delivery and health outcomes, (ii) identify the contextual factors that influence effective integration of DH into healthcare workflows and (iii) highlight key health outcomes that are influenced by the implementation of DH strategies in SSA countries.

The rest of the paper is organized as follows: In section 2, the research methods are briefly outlined, followed by the presentation of the study's results of the data synthesis and analysis in section 3. Section 4 presents a discussion of the findings, and the paper concludes with recommendations of potential future research directions in section 5.

2. Method

To achieve the aim of this paper, a realist review was undertaken. This review synthesized existing literature on the integration of DH technologies into healthcare workflows with the goal of enhancing service delivery and promoting sustainable and equitable health outcomes in SSA countries. The realist approach was employed as an explanatory methodological approach for evidence synthesis (Pawson et al., 2005; Rycroft-Malone et al., 2012). Based on the realist ontology, the realist review posits that the success of programmes such as effective implementation of DH strategies, depends on how mechanisms, contexts, and outcomes interact. Thus, the method centres on Context-Mechanism-Outcome (CMO) configurations to build or refine programme theories or hypothesis about resources a programme provides and how recipients reason about them to produce outcomes (Pawson & Tilley, 1997). In the CMO configuration, *context* is the programme's environment, which can change how mechanisms work, and affect outcomes. Contextual factors can be socio-cultural, technical, or political. *Mechanisms* explain how different conditions and structures create relationships, leading to *events/outcomes* in real and observed worlds (Jagosh et al., 2015; Sayer, 2000). These outcomes can be positive, negative, intended, or unintended, showing the programme's impact.

Within this paper, the realist synthesis enabled the researcher to move beyond simply asking whether the implementation of DH strategies or related ICT health policies facilitate adequate integration of DH technologies into healthcare workflows or not. Essentially, it allowed the examination of the *how, why,* and *under what conditions* the implementation of DH strategies work or not. Specifically, the realist synthesis helped in identifying and articulating casual explanations (theories) for both the regularities and irregularities observed in the implementation of DH strategies. The method also highlights how contextual factors shape what actors in the implementation process can do or cannot do, leading to recurring but only predictable patterns of behaviour, known as demi-regularities (Wong et al., 2010). Thus, a realist synthesis looks for theories that explain these patterns by studying how different situations and actions work together. This helps in understanding specific programme outcomes (Kirst & O' Campo, 2014; Pawson, 2006; Wong et al., 2010). The realist synthesis was conducted using four main steps adapted from Pawson (2006). The application of these steps is discussed in detail in the following sub-sections.

Step 1: Defining the focus and identifying initial programme theories

The focus of the paper was on how effective implementation of DH strategies and related policies enables the integration of DH technologies (such as mHealth, telehealth, and EHR systems) into healthcare workflows to improve service delivery and promote sustainable,

equitable health outcomes in SSA. Guided by the CMO configurations, the synthesis aimed to identify and refine programme theories explaining *what works, for whom, in what contexts, and why*. From a review of scholarly literature and various national DH strategies and related policies, seven initial programme theories were developed (*see* Table 1).

Step 2: Search Strategy

An iterative and purposive search strategy was employed to identify relevant literature published between 2014 and 2025. Searches were conducted in major electronic databases, including Scopus, Pubmed, and Google Scholar. This was complemented by targeted searches of grey literature and policy documents from WHO, ITU, and national DH repositories. Key search terms combined concepts relating to "Digital Health", "Utilisation", "Barriers", "Effectiveness", and "Health outcomes") derived from the paper's objectives. These were combined into detailed search strings, such as "Digital health" OR ("mHealth" OR "eHealth") AND ("use" OR "adoption" OR "integration") AND ("barriers" OR "challenges" OR "obstacles") AND ("causal mechanisms" OR "factors influencing effectiveness" OR "determinants of success") AND ("healthcare outcomes" OR "healthcare delivery") AND ("Sub-Saharan Africa" OR "Africa"), to ensure a broad yet relevant retrieval of literature.

Table 1
Initial programme theories with CMO configurations.

Programme Theory	Description (CMO)
Building institutional commitment through coordinated governance	This theory posits that strong national DH strategies and visible leadership (*context),* fosters governance structures and accountability that convince implementers of sustained commitment *(mechanism)*. This leads to coordinated, streamlined, and scalable adoption of DH technologies *(*outcome*).*
Generating user trust through infrastructure reliability & system interoperability	This theory asserts that reliable infrastructure and interoperability standards provide the technical foundation shared platforms *(context)*. This creates confidence among users that systems will work reliably *(mechanism)*, leading to seamless workflow integration, the elimination of parallel paper systems, and more timely care *(outcome)*.
Fostering sustained adoption through capability-building & peer validation	This theory posits that investing in training, mentoring, and local champions (*context)*builds stakeholders such as healthcare workers' digital capacity and motivates them through recognition (*mechanism).* This results in higher adoption correct usage, and the local capacity to adapt DH tools (*outcome).*
Achieving sustainability through strategic institutionalization	This theory posits that concreate financing plans and procurement mechanisms *(context)* assure healthcare providers that DH systems will be sustained beyond phases *(mechanism)*. This confidence leads to long-term operation and scale-up, preventing tool abandonment *(outcome).*
Creating user ownership through participatory co-design	This theory posits that multi-stakeholder engagement in the design process *(context)* creates user ownership and ensures tools address practical workflows and ethical concerns *(mechanism)*. This results in higher acceptability, a better clinical fit, and more equitable outcomes *(outcome)*
Enabling confident engagement through legal trust architecture	This theory asserts that clear data governance and privacy framework *(context),* build trust and reduce legal risks for providers and patients *(mechanism)*. This increases the uses of digital systems for recording and sharing data, improving data quality and care coordination *(context)*
Ensuring contextual responsiveness through evidence-informed adaptation	This theory posits that built-in monitoring and evaluation frameworks *(context)* provide evidence that enables implementers to iteratively adapt tools *(mechanism).* This results in continuous improvement, better workflow alignment, and a stronger case for continued investment *(outcome).*

Inclusion and exclusion criteria

Studies were included if they: (i) examined the design, implementation, or evaluation of DH strategies, policies, or programmes in SSA countries, (ii) Reported on factors affecting integration, sustainability, or equity in DH technology use in healthcare workflows, and (iii) provided empirical data or interpretive insights relevant to at least one CMO configuration. Editorials, opinion pieces without empirical evidence, articles not written in English or studies conducted outside SSA were excluded unless they provided transferable theoretical insights.

Step 3: Data Extraction and Coding

A structured CMO coding template was developed to guide data extraction (*see supplementary File 1)*. For each included article, data were coded under the following headings:

i. **Context (C):** structural, organizational, and socio-political conditions influencing effective implementation of DH strategies and related policies.
ii. **Mechanism (M):** resources introduced by the programme and participants' reasoning or responses.
iii. **Outcome (O):** observable effects, both intended and unintended, on workflow integration, service delivery, and equity.

The coding process combined deductive and inductive approaches within a retroductive logic framework. Deductive codes, drawn from initial programme theories, provided a structured lens for analysis, while inductive coding captured emergent mechanisms such as trust in digital systems and variations in digital literacy. The author and two other reviewers independently coded the data, resolving differences through consensus. This iterative process enabled the analysis to move beyond surface patterns, uncovering deeper generative conditions. By integrating deductive expectations with inductive insights, the study developed explanatory account of how features DH strategies either support or hinder the integration of technologies into healthcare workflows.

Rigour and quality appraisal

To ensure rigour, each included article was appraised for relevance (*contribution to theory-building*) and rigour (*credibility and trustworthiness of methods*) rather than methodological hierarchy. Triangulation across data types (qualitative and quantitative) enhanced validity. It is important to note that no publication was excluded solely on quality appraisal; rather this step helped identify the strongest evidence to contribute to the synthesis. Reflexive discussions among the review team were used to challenge assumptions and strengthen the explanatory power of final programme theories.

Step 4: Data Synthesis and Refinement of programme theories

The data synthesis unfolded through three iterative stages: *configuration, comparison and consolidation.* In the configuration stage, individual CMOs were constructed from coded evidence drawn within and across the reviewed articles. During the comparison stage, these configurations were examined across different contexts to identify recurrent or divergent mechanisms. Finaly, in the consolidation stage, the CMOs were linked into explanatory pathways, producing refined programme theories that illustrate how specific DH strategy implementation features either enable or impede adequate integration of DH technologies into healthcare workflows. Emergent findings were continuously tested and refined against additional sources through theoretical saturation and retroductive reasoning, consistent with realist logic of inquiry. The data synthesis and refinement of programme theories is further discussed in the **Result of the Data Synthesis and Analysis** section.

3. Result of the data synthesis and analysis

A total of 30 articles published between 2014 and 2025 were included (see Fig. 1), covering twelve (12) SSA countries. These countries included those with robust DH ecosystems such as South Africa, Kenya, Rwanda, Nigeria, Ghana, and Tanzania. These countries have demonstrated notable progress in integrating DH interventions into healthcare workflows, with the aim of enhancing service delivery and health outcomes. Nonetheless, the study also acknowledges the important advances being made in other countries across the region.

DH interventions included mHealth (n=10), telehealth and telemedicine (n=5), EHR (n=3), healthcare information systems (n=3), wearable apps (n-1), AI use ($n = 1$), and DH technology use in general (n=7). Study designs comprised qualitative case studies ($n = 13$), quantitative observations ($n = 4$), systematic reviews ($n = 8$), mixed-methods evaluations ($n = 1$), and cross-sectional case studies (n=4). Data collection methods varied, with studies addressing public healthcare facilities in both rural and urban settings, national populations, community levels, or individual groups. A summary of study characteristics in presented *Table 2 (see supplementary File 2).*

3.1. Refined programme theories (CMOs)

The refined programme theories presented here provide robust explanations of *what works, for whom*, and *in what circumstances* during the implementation of DH strategies and related policies in SSA healthcare systems. The theories explicate the underlying mechanisms (*the reasoning and responses of key stakeholders such as patients, healthcare workers and facility managers*). They show how critical contextual factors trigger, enable, or hinder the identified mechanisms during the DH strategy implementation process. By clarifying pathways linking context, mechanisms, and outcomes, these seven theories explain variable patterns of success and failure, generating actionable insights to guide effective implementation of DH strategies. Thus, rather than listing inputs or interventions, these refined theories expound the reasoning process and behavioural shifts that operationalise DH strategies into practice in facilitating the integration of DH technologies into healthcare workflows. Together, they provide implementers with a comprehensive understanding of not just *what* to do but *how* and *why it works* across diverse SSA contexts.

Theory 1. Building Institutional Commitment Through Coordinated Governance

In SSA, the integration of DH interventions into healthcare workflows is persistently hindered by several challenges including but not limited to infrastructural disparities, limited resources, weak governance, and shortage of skilled professionals, exacerbated further by brain drain phenomenon (Achieng & Ruhode, 2023; Dadzie Ephraim et al., 2024; Ddamba et al., 2025; Maphumulo & Bhengu 2019; Top, Konca, & Sapaz 2020). Despite these complex challenges, this first theory demonstrates how even within such constraints, *institutional commitment* can emerge. It explains how *governance structures*, by creating legitimacy signalling, can change stakeholder assessment regarding the investment value of DH interventions.

CMO configuration statement

In fragmented healthcare systems with multiple disconnected actors competing for limited resources, weak coordination mechanisms, and histories of abandoned DH initiatives **(C)**, the alignment of strong leadership, effective governance and policy coherence creates robust institutional resources. This triggers a cognitive reappraisal in stakeholders, who conclude the intervention is legitimate and has long-term viability, moving beyond a "wait and see" mindset. This cognitive reappraisal activates mechanisms of *confidence in institutional commitment* and *perceived strategic legitimacy* **(M)**, which leads to proactive engagement, manifested through committed resources and collaboration, which drives coordinated implementation and sustained DH integration **(O)**.

In this theory, the governance mechanism operates through legitimacy signalling that resolves stakeholder uncertainty regarding institutional commitment. In resource-constrained healthcare institutions across SSA, where skepticism persists due to past DH failures or

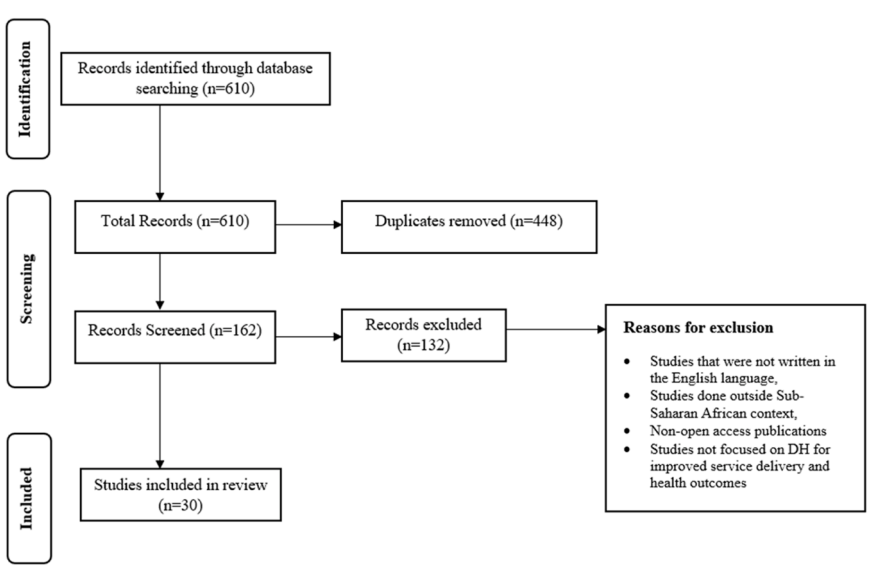

Fig. 1. Flowchart of search results.

abandonment, governance mechanisms build credibility by signalling legitimacy (Bakibinga et al., 2020; Galani et al., 2021; Ogugua et al., 2024; Opoku et al., 2017). This occur when three elements align: First, *Strong leadership* demonstrates commitment through tangible actions like budgeting and public championing (Agyepong et al., 2018; Mboera et al., 2021; Mbunge et al., 2022). Second, *effective governance* translates this vision into reality with clear accountability structures and transparent rules (Kapepo et al., 2024; Mbunge & Sibiya, 2024). Finally, *policy coherence* complements leadership and governance by embedding the initiative within broader national plans, proving it is a strategic priority, not a fleeting pilot. This combined signalling resolves uncertainty, transforming skepticism into engaged participation (Aboye et al., 2024; African Development Bank, 2022).

Together these elements create a "credible cascade" where each dimension reinforces the others. Strategically aligned, they trigger a cognitive shift in stakeholders, who come to see DH interventions not as "another project to manage" but as an institutional transformation (Olufadewa et al., 2024; WHO Regional Office for Africa, 2024). Conversely, misalignment triggers defensive, risk-averse reasoning. Leadership rhetoric without governance is seen as "performative theatre", and policy without leadership support as a mere "administrative formality". In these conditions, rational actors in the implementation process of DH strategies and related policies, minimize their commitment, creating self-fulfilling failures that erode institutional credibility and reinforce pre-existing skepticism (Alunyu et al., 2024; Ogugua et al., 2024).

Theory 2. Generating User Trust Through Infrastructural Reliability & System Interoperability

This second refined theory demonstrates how reliable infrastructure creates conditions for users to trust and integrate digital tools into practice. Therefore, *Infrastructural reliability* and *system interoperability* form one of the non-negotiable technical foundations for any effective implementation of a DH strategy initiative (Olufadewa et al., 2024; WHO Regional Office for Africa, 2024). Historically, weak governance structures in many SSA countries have led to considerable variation in DH infrastructure and interoperability (Mremi et al., 2022; Ndlovu et al., 2021). The implication of this is the inability of healthcare institutions to adequately integrate DH interventions into healthcare workflows. For stakeholders such as healthcare workers and patients who are accustomed to unreliable digital infrastructure, the consistent experience of robust, interoperable systems catalyses a fundamental shift in reasoning (Mindu et al., 2023; Onsongo et al., 2023). The perception moves from seeing digital tools as a burden to recognizing them as a capability enhancer. This hard-earned trust, built through seamless functionality, fosters the progressive integration of DH into healthcare workflows, ultimately producing improved care continuity and data-driven decision-making.

CMO configuration statement

In healthcare facilities across SSA, unreliable internet connectivity, unstable power supply, fragmented digital systems, and prior negative experience with failed implementations **(C)**, fosters skepticism towards DH. Where robust infrastructure is introduced, stable power supply, reliable connectivity, interoperable platforms, and responsive technical support, healthcare workers encounter consistent positive experience, such as seamless data exchange, reliable saving, and rapid system response. These repeated interactions recalibrate expectations, shifting digital tools from perceived burdens to dependable assets and activating mechanisms of *trust* and *professional empowerment* **(M)**. As trust consolidates, workers integrate digital systems into routine workflows, reduce reliance on paper backups, and extend use beyond compliance to clinical decision-making. This behavioural shift supports advocacy among peers and produces system-level outcomes, including improved continuity of care, enhanced data quality for planning, and more equitable service delivery through innovations such as mHealth and telehealth **(O)**.

The mechanism of *trust* in DH systems and technologies develops incrementally through accumulated experience. Initial positive encounters such as systems functioning reliably during busy clinical shifts or health-related data remaining accessible when needed, create openings for healthcare workers to reconsider prior skepticism. When these experiences are consistently reinforced, cognitive models are updated and digital tools come to be seen as genuinely reliable and integral to workflow planning (Dodoo et al., 2021; Sagaro et al., 2020). Interoperability further amplifies this process by enabling seamless data exchange across facilities. In such contexts, healthcare workers perceive themselves as part of a connected system rather than isolated actors, fostering both digital trust and a sense of professional empowerment (Ngongo et al., 2019; Zayyad & Toycan, 2018). In contrast, when infrastructure is unreliable, a negative cycle develops. Frequent system failures lead to growing distrust and dissatisfaction, which reduces usage, strengthens dependence on parallel paper-based systems, and results in continued underinvestment in DH (Agboryah et al., 2024; Bogale et al., 2023; Odekunle et al., 2017).

Theory 3. Fostering Sustained Adoption Through Capability-Building & Peer Validation

The human resource dimension is a critical factor in translating DH strategy initiatives into tangible improvements in healthcare delivery (Atukunda et al., 2021; Manyazewal et al., 2021). This refined theory explains how sustained adoption of DH technologies emerges through two complementary mechanisms: *capability-building,* which enhances individual self-efficacy and *peer validation,* which reinforces social legitimacy. The theory emphasizes that even when reliable infrastructure and strong institutional support are in place, DH tools often remain underutilized if users lack confidence in their own ability to use them or if the tools are not endorsed by trusted local champions. Without capacity-building mechanisms such as continuous and targeted training, even the well-designed and well-intentioned DH interventions risk creating greater burdens than benefits.

CMO configuration statement

Across many healthcare facilities in SSA where workforce shortage, high staff turnover, competing clinical demands, and significant digital literacy gaps, especially among senior clinicians and rural practitioners, skepticism towards DH interventions is often heightened. Such skepticism is not only rooted in present challenges but also shaped by institutional memory of past implementation failures, which fosters a culture of caution and resistance **(C)**. Such dynamics highlight the importance of addressing both technical and human resource dimensions in DH adoption and use. Without deliberate strategies to build digital competencies, generate the social proof needed to support change, foster trust, and demonstrate tangible clinical value, even well-designed DH interventions risk being sidelined within routine practice. To mitigate these risks, it is therefore critical that continuous, hands-on, and context-specific training, coupled with the presence of visible champions, deliver both the technical competencies and the social proof required to embed change. Continuous training builds competence and confidence, while observing respected local champions integrate DH tools affirms their value. Together, these mechanisms of skill development and peer modelling foster *self-efficacy* and *trust,* shifting healthcare workers from avoidance to proactive engagement **(M)**. As confidence grows, staff begin to use digital tools meaningfully for clinical tasks, troubleshoot problems independently, mentor colleagues, and adapt systems to local workflows. The result is sustained adoption that endures despite staff turnover, more efficient clinical practice, and progressive innovation as empowered users propose contextually relevant improvements **(O)**.

Adoption of DH tools in clinical settings proceed through two interdependent pathways: *self-efficacy formation* and *social validation*. Digital anxiety is common, particularly among senior clinicians whose professional identifies were formed without digital technologies, undermining willingness to engage when tools are available. Well-designed, continuous training that uses graded mastery experiences (simple, low-stakes tasks progressing to complex scenarios) and

supportive feedback normalizing mistakes builds capability and confidence. As healthcare workers successfully complete tasks, they update their beliefs and sustain gains through ongoing practice rather than one-off sessions (Ndayizigamiye et al., 2018; Petersen et al., 2019; Asah & Kaasbøll, 2023). Alongside capability, legitimacy is essential. Local champions, respected clinicians and managers with unquestioned clinical credibility, provide social proof that DH tools serve care quality and efficiency rather than merely administrative compliance. When healthcare workers observe these Fig.s consistently using DH tools in busy workflows, they infer genuine value and feel permission to adopt. Champions extend formal training through informal mentoring: translating abstract functions into concrete workflows, sharing workarounds, and normalizing struggle in context (Achieng & Ruhode, 2023; Karamagi et al., 2022; Mremi et al., 2022).

These pathways are synergistic and necessary. Training without champions without training can generate aspirational models that leave peers feeling inadequate. Together, they create environments in which healthcare workers feel both competent and socially supported, enabling proactive integration, durable use despite turnover, and progressive innovation as confident users suggest context-specific improvements (Alunyu et al., 2024; Kesse-Tachi et al., 2019). A contrastive pattern underscores this logic: one-off, theoretical training disconnected from real workflows heightens anxiety under clinical pressure, while absent or externally imposed champions lack local legitimacy, leaving tools socially illegitimate despite technical quality (Achieng & Ogundaini, 2022; Alunyu et al., 2024; Asah & Kaasbøll, 2023; Furusa & Coleman, 2018; Karamagi et al., 2022; Mremi et al., 2022; Ndayizigamiye et al., 2018).

Theory 4. Achieving Sustainability Through Strategic Institutionalization

Across SSA, promising DH innovations often demonstrate success in pilot projects but struggle to scale or sustain once initial funding ends (Swartz et al., 2021). This pattern reflects deeper systemic challenges, including limited institutional ownership, inadequate policy integration, weak monitoring capacity, and insufficient DH leadership, which collectively undermines long-term adoption and sustainability (Ogundaini & Mlitwa, 2024). Many healthcare facilities adopt new DH tools and systems with apparent commitment while projects are externally funded, yet these implementations often deteriorate or are abandoned once projects funding ends, an outcome consistently documented across region-wide reviews of DH interventions (Owoyemi et al., 2022; Swartz et al., 2021). This refined theory explains how *strategic institutionalization* breaks this cycle by shifting decision-makers to conceptualize DH investment: from temporary projects requiring external support to permanent infrastructure requiring domestic stewardship. The theory demonstrates that long-term success of DH integration id fundamentally contingent on robust foundation of financial planning, strategic procurement, and deliberate sustainability frameworks.

CMO configuration statement

In SSA healthcare systems, the long-term sustainability and scalability of DH initiatives are often hindered by a combination of factors. They are typically project-based, heavily dependent on donor funding, fragmented, and hindered by weak procurement systems, while the departure of donor support staff leads to a loss of institutional memory **(C)**. Within this context, deliberately integrating DH strategy initiatives into institutional structures, such as national budgets, procurement frameworks that prioritize interoperability, and clear sustainability plans, is critical to shifting perceptions. This process reframes DH initiatives from temporary, donor-funded projects into core, long-term health infrastructure. As a result, decision-makers are prompted to focus on recurrent costs, interoperability, and a credible vision for continuity beyond pilot phases, treating DH as an integrated ecosystem rather than a collection of fragmented tools. Together, these mechanisms foster *institutional ownership* and *stewardship,* replacing short-term thinking with a long-term investment mindset **(M)**. This leads decision-makers to make fundamentally different choices, allocating local resources to operational sustainability, prioritizing interoperability and upgrade pathways when procuring systems, investing in building local technical capacity, and creating institutional mechanisms that outlive individual projects. These actions produce genuine institutionalization where DH interventions become integrated into healthcare system operations, outliving donor funding cycles, leadership changes, and implementation **(O)**.

In healthcare systems across SSA reliant on donor funding, decision-makers often adopt a rationally adaptive "project mindset" shaped by the typically short-term and discontinuous nature of DH initiatives (Binyaruka et al., 2023). This framework encourages prioritizing immediate deployment over long-term sustainability, leading to behaviours such as favouring temporary consultants over training existing staff, procuring non-interoperable systems to minimise upfront costs, and focusing on high-visibility pilot projects rather than systemic rollouts (Binyaruka et al., 2023; Ilesanmi & Afolabi, 2022; Scott & Mars, 2015). A crucial cognitive shift occurs, however, when DH is deliberately embedded within domestic institutional structures. Integrating DH into multi-year national budgets and strategic procurement frameworks fundamentally alters the planning horizon. This institutionalization reframes the guiding question from "what is cheapest now?" to "what offers sustained value over time?", thereby enabling long-term investments in capacity and interoperable infrastructure (Scott & Mars, 2015; WHO, 2016).

This reframing marks the critical transition from viewing DH as a transient project to recognizing it as enduring health infrastructure. Without it, systems remain caught in a self-reinforcing cycle of dependency. Policymakers, uncertain of future funding, withhold domestic investment, and when donor support ends and systems collapse, their initial skepticism is validated, further entrenching the "pilot trap."

Theory 5. Creating User Ownership Through Participatory Co-Design

One of the most critical elements in the implementation of DH strategies across healthcare systems is stakeholder engagement and co-design. This refined theory demonstrates how effective implementation of DH strategy initiatives is fundamentally a socio-technical process, driven by meaningful stakeholder engagement and co-design. In many SSA healthcare systems, the integration of DH technologies into healthcare workflows has been negatively affected by limited stakeholder engagement and insufficient co-design processes, which undermine contextual relevance, user adoption, and long-term sustainability (Mmbali & Chillo 2017; Ogundaini & Mlitwa, 2024; Olufadewa et al., 2024; WHO, 2021). DH strategy initiatives often fail when designed in a top-down manner, overlooking the invaluable insights of end-users' stakeholders such as healthcare professionals, patients, and community leaders. The result is often a poor fit with local workflows, leading to low adoption and limited systemic impact (Ogundaini & Mlitwa, 2024). This theory explains how participatory processes generate *psychological ownership* by giving users agency in shaping technology to their needs.

CMO configuration statement

In many healthcare systems across SSA, DH implementation has historically been designed top-down by external experts without meaningful user input, creating tools misaligned with local workflows, infrastructure constraints (limited connectivity, low spec devices), cultural practices, and actual user needs, resulting in low adoption, workarounds undermining system integrity, and user frustration **(C)**, In this context, participatory co-design emerges as a more effective approach that shifts power to end-users by granting them authentic decision-making authority (agency). When the design also accounts for real-world constraints (contextual fit) and is developed collectively (social embeddedness). These reasoning pathways, agency, contextual fit, and community, activate powerful mechanisms of *psychological ownership, perceived relevance*, and *collective efficacy* **(M)**. The result is a dramatic transformation in behaviour. Users shift from passive compliance or

active resistance to proactive engagement. They willingly adopt the tools, troubleshoot problems constructively, advocate for the systems with skeptical colleagues, and iteratively suggest improvements. This process cultivates a robust ecosystem characterized by sustained adoption, high-quality data born from user investment, and DH systems that evolve and improve organically over time through user feedback **(O)**.

In many SSA healthcare systems, the historical disconnect between technological design and the contextual realities has created a vicious cycle. DH tools that are misaligned with local contexts not only see low usage but can also inadvertently increase the workload for already overburdened healthcare workers, leading to digital burnout and resistance to future initiatives (Akanbi et al., 2012; Medhanyie et al., 2015; Oleribe et al., 2019). Furthermore, a lack of co-design often means that solutions fail to account for critical infrastructural constraints, such as unreliable internet connectivity, limited digital literacy, or the use of low-specification mobile devices (Scott & Mars, 2015). Consequently, even well-intentioned DH initiatives can exacerbate existing inequities, diverting scarce resource toward "white elephant" projects that fail to deliver tangible health outcomes and eroding trust in digital solutions (Dadzie Ephraim et al., 2024; Swahn et al., 2024).

From a realist perspective, the mechanism underpinning stakeholder engagement and co-design lies in the provision of participatory resources that activate trust, ownership, and collective efficacy. When users perceive their contributions as shaping technology, they are more likely to internalise it as a solution to their own needs rather than an imposed system. This shifts reasoning from compliance to commitment, fostering sustained engagement and a culture of innovation. Literature indicates that healthcare workers in healthcare institutions are positioned as co-creators, they translate perceptions of ownership into higher adoption, improved data capturing resulting in high data quality, and greater user satisfaction (Akanbi et al., 2012; Olu et al., 2019). Yet co-design is not automatically inclusive. It can reproduce inequities if marginalised groups are excluded from deliberative spaces. Authentic stakeholders engagement must therefore amplify diverse voices, ensuring representativeness across gender, professionalism, and geographic line. When such conditions are met, co-design transforms stakeholders from passive recipients into active partners, generating mechanism of legitimacy and accountability that extend beyond implementation. Ultimately, participatory processes embed DH interventions within social and organizational contexts, ensuring they are not only technically viable but socially sustainable.

Theory 6. Enabling Confident Engagement Through Legal Trust Architecture

In many countries across SSA, DH tools are often introduced into fragile regulatory environments where data protection laws are inconsistently enforced, and institutional accountability is weak (Awosiku et al., 2025; Ogundaini & Mlitwa, 2024). This has created uncertainty among end-users of these tools who may be reluctant to engage with technology that appear to compromise privacy or impose additional risks (Olufadewa et al., 2024). This refined theory explains *how legal trust architecture,* scaffolding of laws, regulations, enforcement mechanisms, and redress systems, resolves this problem by transforming unmanageable risk into governed, accountable risk, enabling confident participation rather than defensive withdrawal.

CMO configuration statement

In healthcare systems where a combination of factors undermines data security such as weak or absent data protection laws, inconsistent enforcement of existing regulations, histories of data breaches or unauthorized access without consequences, and a lack of transparent data governance creates uncertainty. These issues are particularly acute in contexts with low institutional trust, especially among marginalized populations who have previously experienced discrimination **(C)**. In such contexts, a *comprehensive legal trust architecture* addresses vulnerabilities through four pillars: (i) clear legislation defining individual rights and institutional responsibilities; (ii) robust enforcement mechanisms that empower regulatory authorities to investigate and impose meaningful penalties for violations; (iii) accessible redress that lower barriers to lodging complaints; and (iv) transparency requirements that make data practices visible and auditable. Together, these resources reshape how patients and providers reason about health-related data use. Explicit laws signal that rights are recognized, enforcement ensure misuse carries consequences, and redress mechanisms provide avenues for remedy. These shifts foster *perceived safety, reduce risk assessment*, and *empower individuals through legal recognition* **(M)**. The outcomes of this reasoning are tangible. Healthcare workers, reassured by institutional safeguards, record sensitive health information accurately, while patients, perceiving reduced risks, disclose complete health histories, rather than withholding stigmatized details. Both sustain engagement with DH systems and report violations when they occur, enabling system learning. This produces improved data quality, (complete, accurate information enabling better care), strengthened system legitimacy (users view SH as trustworthy), and a DH ecosystem where privacy and innovation coexist **(O)**.

In realist terms, these frameworks supply the resources (clear rules, enforceable protections, transparent governance) that activate reasoning (trust, ownership, and willingness to participate). For example, Kenya's Data Protection Act (2019) and South Africa's Protection of Personal Information Act (POPIA) have been cited as critical steps in embedding DH within a rights-based framework that reassures users about the safety of their data (Olufadewa et al., 2024; Staunton et al., 2021). However, such legal frameworks are absent in many other SSA countries. DH initiatives often face resistance, low uptake, and poor data quality (Musa et al., 2023). Conversely, when healthcare workers and patients perceive that their rights are safeguarded, they are more likely to share accurate information, adopt new tools, and sustain their use overtime (Aboye et al., 2024). This shift in reasoning, from skepticism to confidence, translates into outcomes such as improved data reliability, enhanced legitimacy of DH initiatives, and reduced risk of "pilot fatigue" where projects are abandoned after enthusiasm wanes.

The presence of legal frameworks alone is not sufficient. Trust architectures must also be inclusive, ensuring that protections extend to marginalised groups who are often excluded from decision-making spaces. Without deliberate attention to equity, legal frameworks risk reproducing existing inequities, thereby undermining the very mechanisms of trust and legitimacy they are meant to activate (Olufadewa et al., 2024). National DH strategies across SSA reveal significant variability in this regard, with some countries embedding strong provisions for accountability and inclusivity, while others remain weak or outdated (Ogundaini & Mlitwa, 2024; Olufadewa et al., 2024). Ultimately, this refined theory suggests that trust architecture and legal frameworks are not peripheral but central to the success of DH interventions. By embedding DH within transparent, enforceable, and inclusive governance systems, these frameworks transform stakeholders from passive recipients into active partners. In doing so, they generate mechanisms of legitimacy and accountability that extend beyond implementation, ensuring that DH interventions are not only technically viable, but also socially sustainable (Awosiku et al., 2025; Olu et al., 2019).

Theory 7. Ensuring Contextual Responsiveness Through Evidence-Informed Adaptation

Monitoring and evaluation (M&E) remain a critical yet underdeveloped component in the implementation of DH strategies across SSA. While the use of DH interventions ranging from mHealth apps to EHRs have expanded rapidly, their sustainability is often undermined by fragmented and inconsistent M&E frameworks (Karamagi et al., 2022; Otorkpa et al., 2024). The implementation of such interventions often encounters persistent challenges including workflows misalignment with actual practice, users' resistance, technical problems surfacing only at scale, and inequities hidden during pilot phases. Despite these realities, most implementations remain static, designed once and executed rigidly, even as discrepancies with contextual realities become evident.

This resulting "implementation drift", the widening gap between planned design and lived practice, undermines initiatives and leads to failure, even when the underlying technology itself is robust. This refined theory, explains how *evidence-informed adaptation,* the capacity to learn from implementation and adjust course, closes the gap between planned and actual implementation through continuous organizational learning.

CMO configuration statement

In complex and unpredictable healthcare environments characterized by diverse facility contexts (urban/rural, well-resourced/under-resourced), implementation challenges often emerge at scale, when infrastructure variability, workflow diversity, and unanticipated user behaviour become visible. Resource constraints further heighten the stakes: limited budgets cannot absorb failed investments, and staff time too scarce to allow repeated cycles of retraining. Within such contexts, rigid implementation approaches are particularly vulnerable to failure **(C).** A more resilient pathway lies in monitoring and evaluation (M&E) systems designed for learning rather than accountability, paired with organizational structures that empower implementers to act on evidence. These systems provide real-time insight into usage, feedback, performance, inequities, and unintended consequences, while authority structures enable timely adaptation across facility, district, and national levels. When implementers receive feedback, such as lower adoption rates in rural clinics due to connectivity barriers, they can reframe problems as structural rather than behavioural and with authority to adapt, enact solutions immediately. This combination fosters *organisational sense-making, empowered problem solving*, and a *continuous learning orientation*, shifting implementation from rigid plan execution to adaptive problem-solving **(M)**. The outcomes are clear: facility staff adapt workflows to local practices, district managers reallocate resources, and national teams revise guidelines based on field experience. Iterative testing and refinement align implementation with diverse realities, making it more resilient, equitable, and efficient. Rather than drifting toward failure, systems evolve toward effectiveness, showing how adaptive learning infrastructures can transform digital health implementation **(O)**.

Many DH initiatives in SSA countries prioritize pilot projects without embedding systematic evaluation mechanisms, resulting in limited evidence on scalability and long-term impact (Akanbi et al., 2012). Furthermore, M&E processes are frequently donor-driven, emphasizing reporting adaptive learning, which constrains their ability to inform policy and strengthen healthcare systems (Nabyonga-Orem et al., 2021). In unpredictable and often resource-constrained healthcare systems in SSA, the mechanism of M&E is vital, as it provides the empirical basis for real-time course correction and fosters a nuanced, empathetic understanding of the user experience. Literature highlights the need for context-sensitive, participatory M&E approaches that not only track progress but also guide decision-making and resource allocation (Otorkpa et al., 2024). Strengthening M&E capacity is therefore essential for ensuring that DH strategies contribute meaningfully to resilient and equitable healthcare systems in SSA.

Adaptive learning then acts as the responsive brain, depending on both cognitive resources (understanding what is happening and why) and organizational resources (the authority and mean to act). This pathway unfolds through four interlinked phases. First, **evidence generation** relies on learning-oriented M&E that produces rapid feedback and encourages honest, non-punitive input from frontline users (Nabyonga-Orem et al., 2021; Otorkpa et al., 2024). Second, **sense--making** transforms observed patterns into causal insights, reframing issues such as low rural adoption not as resistance but as structural constraints that can be addressed (Karamagi et al., 2022; Otorkpa et al., 2024). Third, **empowered adaptation** occurs when distributed authority allows implementers at all levels to act within defined guardrails, motivating them to invest in problem analysis because they can act on what they learn (Kipruto et al., 2022). Finally, **iterative refinement** normalizes "productive failure," testing adaptations in limited settings, scaling what works, and learning from what does not (Otorkpa et al., 2024). By contrast, when M&E is absent or reduced to compliance, implementers neither understand problems nor have the authority to address them. Rigid hierarchies delay adaptation, resources continue to flow into ineffective strategies, and implementation drift deepens until problems become entrenched and costly to reverse (Akanbi et al., 2012; Karamagi et al., 2022).

3.2. Synthesis across the refined theories

The realist synthesis of evidence, conducted to understand *how, for whom,* and *under what circumstances* the implementation of DH strategies enable the integration of DH technologies is successful, refined seven interconnected theories. A central finding of this review is that the identified mechanisms do not function in isolation but instead operate synergistically. Achieving effective, equitable, and sustainable integration of DH technologies within SSA healthcare systems depends on the simultaneous activation of multiple mechanisms, each of which addresses distinct yet interdependent contextual barriers and enablers that shape the integration of DH into healthcare workflows. The synthesis organized the seven theories into three interrelated functional categories: *foundational enabling conditions, technical and human enablers*, and *adaptive mechanisms*.

3.2.1. Foundational enabling conditions: establishing the institutional architecture

Under this functional category, the synthesis categorised three theories that form the foundational enabling conditions for effective implementation of a DH strategy or related policy. While these mechanisms do not directly drive the adoption and use of DH technologies, the evidence from the synthesis indicate that they function as critical antecedents by establishing the institutional architecture, legitimacy, permanence, and safety upon which all subsequent DH strategy implementation process rely.

Theory 1, *institutional commitment through coordinated governance,* was identified as the primary authorising mechanism. It suggests that sustained governance commitment generates the political and legitimacy necessary to mobilize resources, mandate interoperability across DH systems, and secure the engagement of key stakeholders. In doing so, it provides DH initiatives with the institutional stability required to endure beyond individual political cycles. **Theory 4**, *sustainability through strategic institutionalization,* advances our understanding of the conditions necessary for the long-term success of DH in SSA healthcare systems. The evidence from the reviewed articles indicates that, in the absence of this mechanism, DH initiative risk remaining short-lived projects. Strategic institutionalization, achieved through dedicated budgets, permanent staffing, and integration into core healthcare systems functions, serves as the critical process that transforms DH from a temporary intervention into a permanent and reliable component of health infrastructure. **Theory 6**, *confident engagement through legal trust architecture,* emerged as a pivotal non-technical foundational mechanism. It highlights the role of a robust, rights-based legal and ethical framework in fostering trust. Such a framework is essential for cultivating confidence among both patients and healthcare workers, reducing likelihood of disengagement driven by concerns over data privacy and misuse, and ensuring that users feel psychologically safe to participate in DH systems. Together these foundational enabling conditions establish the *legitimacy, stability*, and *trust* upon which the subsequent technical and human enablers build to drive the effective adoption of DH in SSA.

3.2.2. Technical and Human enablers: creating operational capacity

Building on the institutional foundation factors, the two theories were identified that address the technical and human enablers essential for the practical, day-to-day implementation of DH. These mechanisms generate the operational capacity needed to translate high-level DH strategies and policies into effective end-user practice.

Theory 2, *user trust through infrastructural reliability*, highlights the material conditions underpinning DH use. It emphasizes that dependable infrastructure such as stable electricity, reliable connectivity, and accessible technical support, functions as a trust-building mechanism. When systems operate consistently within local contexts, they are perceived as credible and useful tools rather than additional burdens, thereby reducing the risk of user abandonment. **Theory 3**, *sustained adoption through capability-building and peer validation*, extends beyond conventional training approaches. The evidence refines this theory by demonstrating that adoption is a socio-technical process: while users must acquire the necessary skills, long-term engagement is equally shaped by social and peer validation. This mechanism ensures that users develop not only the technical competence but also the supportive networks required to use DH technologies effectively and to navigate challenges as they arise. Together, infrastructural reliability and capability-building with peer validation provide the technical and human enablers that convert high-level DH strategies into *credible, trusted*, and *sustainable* adopted practices within everyday healthcare settings.

3.2.3. Adaptive Mechanisms: ensuring long-term evolutionary capacity

The final category of theories highlights the adaptive mechanisms necessary to ensure long-term relevance, contextual fit, and system evolution. In complex healthcare environments, no initial design is flawless; these mechanisms provide the adaptive capacity for DH implementations to learn, adjust, and respond to emerging realities.

Theory 5, *user ownership through participatory co-design*, emphasizes the importance of aligning DH systems with local needs and workflows. By actively involving end-users in both design and iterative refinement, this mechanism fosters a sense of ownership and investment that is critical for social acceptance and sustained use. **Theory 7**, *contextual responsiveness through evidence-informed adaptation*, highlights the role of structured learning processes. It demonstrates how systems that systematically gather and act upon implementation data and user feedback can adapt to evolving challenges. In doing so, this mechanism mitigates the "project drift" often associated with rigid, top-down approaches, enabling interventions to remain relevant and resilient over time.

A key finding of this synthesis is the synergistic interaction among the theoretical mechanisms, which often generate reinforcing cycles. No single mechanism is sufficient on its own, instead, coordinated governance (Theory 1) consistently emerges as the primary enabler, creating the political space and resource flows necessary for infrastructure reliability (Theory 2), legal frameworks (Theory 6), and institutionalization pathways (Theory 4). Operational capacity is further strengthened through the interplay of infrastructural reliability (Theory 2) and capability-building (Theory 3): training on dependable systems fosters genuine competence, while skilled users, in turn, maximize the value of reliable infrastructure. The adaptive mechanisms also demonstrate interdependence. Participatory co-design (Theory 5) cultivates ownership and collaboration, which enhance evidence-informed adaptation (Theory 7), while legal trust architecture (Theory 6) provides the psychological safety for authentic engagement. Finally, strategic institutionalization (Theory 4) safeguards this adaptive capacity by ensuring stable resources and dedicated teams, allowing learning and adaptation to accumulate and inform long-term strategy beyond short-term project cycles.

Fig. 2 depicts the proposed conceptual framework for effective implementation of DH strategies. The conceptual framework highlights how the seven refined theories interact as an integrated system operating across three structural layers and three temporal phases, all within the challenging contextual realities of SSA healthcare systems. The framework suggests that the DH strategy implementation process begins with the institutional foundation layer (Theories 1, 4, 6), which establishes essential enabling conditions through governance, institutionalization, and legal trust during Phase 1: Foundation Building. Next, the operational capacity layer (Theories 2, 3) builds functional implementation capacity in Phase 2, Operational Deployment. Finally, the adaptive evolution layer (Theories 5, 7) fosters continuous improvement in Phase 3: adaptive evolution. Implementation outcomes then reach a critical juncture, diverging into either a successful path, creating

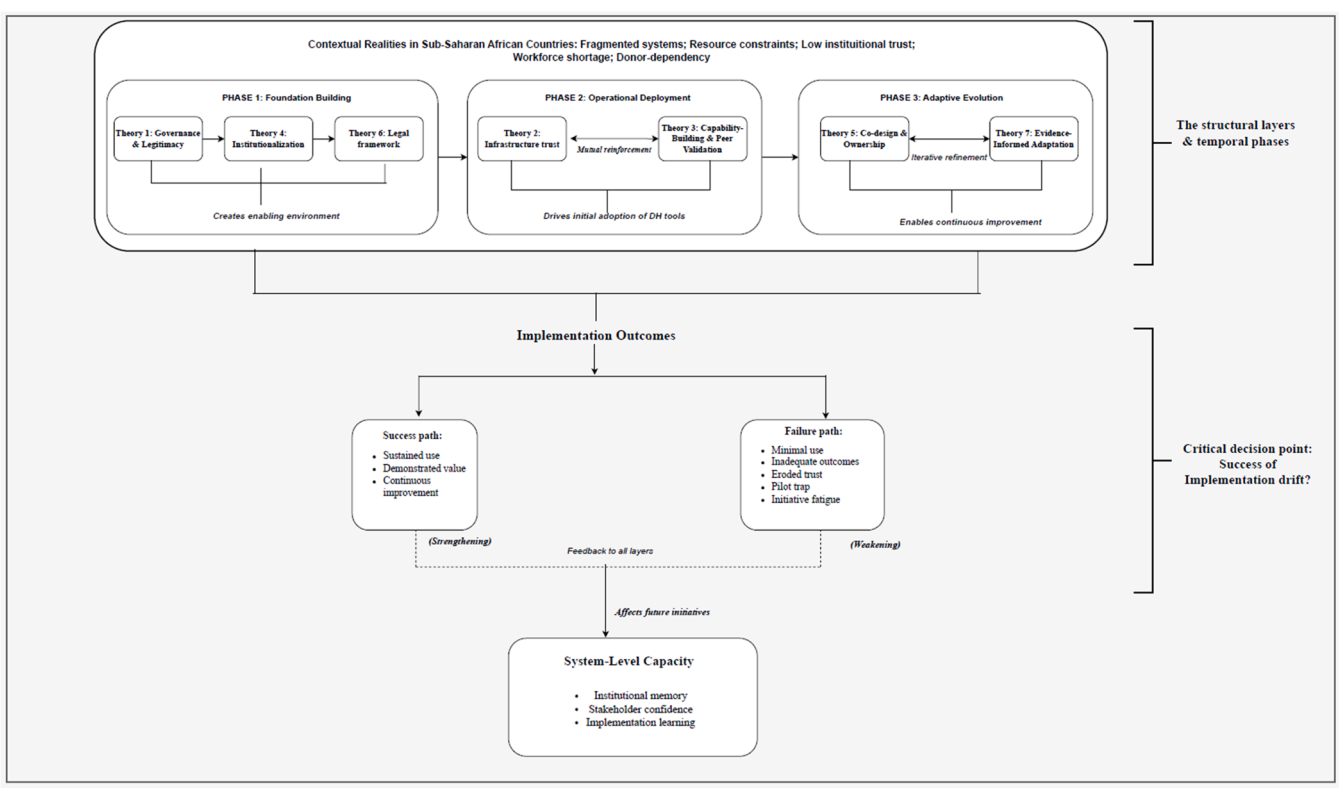

Fig. 2. Conceptual Framework: The interconnected Mechanisms of DH strategy implementation in SSA healthcare systems.

virtuous cycles of sustained use, or failure paths, leading to vicious cycles of initiative fatigue. The feedback loops, illustrated by dashed arrows, demonstrate how these outcomes either strengthen or weaken the entire system's capacity for future initiatives. Ultimately, success requires coordinated investment across all layers and phases, as small and siloed interventions typically fail due to the system's deep interdependencies.

4. Conclusion and recommendations

This realist synthesis was conducted to explain how, why, and under what conditions the implementation of national DH strategies lead to enhanced health outcomes in SSA. The findings moved beyond a simple evaluation of inputs and outputs to uncover the underlying causal mechanisms. The central finding of the review is that effective, sustainable, and equitable DH integration is not the result oof any single intervention, but rather the product of a complex, synergistic ecosystem of seven interconnected mechanisms. The synthesis refined and clustered these mechanisms into three functional and interdependent categories. The first category was ***foundational enabling conditions*** (coordinated governance, strategic institutionalization, and legal trust architecture) that establish the institutional legitimacy, permanence, and psychological safety necessary for any DH initiative to take root. The second was ***technical and human enablers*** (infrastructure reliability and capability-building with peer-validation) that create the operational capacity and socio-technical trust required for frontline adoption and use. Third category was the ***adaptive mechanism*** (participatory co-design and evidence-informed adaptation) that ensure DH interventions are fit-for-purpose and possess the evolutionary capacity to respond to complex, emergent realities over long term.

A critical insight from this review is that these mechanisms are mutually reinforcing. Governance enables infrastructure investment, legal trust enables authentic co-design, and strategic institutionalization protects the capacity for adaptive learning. The persistent challenge of pilot project and the frequent failure of DH interventions to scale can be directly attributed to a failure to activate this entire ecosystem of mechanisms simultaneously. This review, therefore, provides a significant contribution by shifting the focus of policymakers and implementer, from an interventionist question of "what technology should be deployed?" to a realist question of "what foundational, enabling, and adaptive mechanisms must be activated for any technology to succeed?"

Recommendations

Drawing on the seven refined programme theories, this paper offers four key recommendations for policymakers, ministries of health, and implementing partners in SSA.

Adopt a phased, synergistic implementation model: DH strategies should be operationalized as integrated systems rather than fragmented workstreams. Implementation must be sequenced. First securing foundational conditions (governance, institutionalization, and legal trust: Theories 1, 4, 6), then building technical and human enablers in tandem (infrastructure and capability: Theories 2, 3), and embedding adaptive mechanisms from the outset (co-design and learning: Theories 5, 7).

Shift investment from projects to infrastructure: To break the cycle of donor-dependent pilots, governments should prioritize institutionalization (Theory 4) by earmarking domestic funds for DH operations, staffing, and maintenance. Strengthening governance (Theory 1) and resourcing legal trust frameworks (Theory 6) are equally critical to ensure policy coherence, regulatory enforcement, and public confidence.

Treat human capacity as critical infrastructure: Adoption is a socio-technical process (Theory 3). Investments must extend beyond hardware to continuous, workflow-based training, mentorship, and peer validation through local champions who foster trust and overcome resistance.

Mandate co-design and adaptive learning: To prevent "implementation drift," co-design (Theory 5) should be a procurement requirement, ensuring DH tools reflect end-user needs. Monitoring and evaluation (Theory 7) must be designed for adaptive learning, providing real-time feedback that empowers local implementers to make evidence-informed adjustments.

This realist synthesis has provided a robust explanatory model. The next step is to test this model prospectively. Future implementation research should be designed to explicitly test the synergistic application of these seven mechanisms, evaluating how their integration influences implementation success, cost-effectiveness, and, ultimately, measurable health outcomes.

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Mourine S. Achieng: Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

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Supplementary materials

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